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Scalable multiscale echo state reservoirs with dendritic readouts for intelligent earthquake early warning systems

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ABSTRACT

Accurate and prompt prediction of earthquake magnitude using the early few seconds of earthquake wave is a critical objective of onsite earthquake early warning systems. However, it remains challenging because of the highly nonlinear and complex nature of seismic signals. Recent advancements in deep sequence-to-sequence learning have demonstrated strong performance in this domain. However, their multi-million trainable parameters require high-end computational systems. On the other hand, conventional early warning approaches primarily rely on regression-based methods that are computationally efficient but unable to model the nonlinear dynamics inherent in seismic signals. To address this, we propose an earthquake echo state network with dendrites, a lightweight sequence to scalar architecture based on (i) multiscale and multi-layer echo state networks and (ii) enhancement with dendritic neural network layers to achieve non-linearity at reservoir read-out. Echo state networks are a type of recurrent neural network characterized by a fixed, sparsely connected dynamic reservoir. Quantitative analysis reveals that the framework enables efficient modeling of seismic time series with training times of 7.80, 6.71, 10.55, 9.32, and 10.99 s for 2, 3, 4, 5, and 6 s of primary waveform, respectively. The model is evaluated using 36196 three-component seismic records with primary wave windows ranging from 2 to 6 s. A comparative analysis with state-of-the-art methods demonstrates that the proposed method delivers superior performance in terms of both prediction error and computational efficiency, making it a practical solution for onsite early warning deployment on low-resource systems.

1. Introduction

The impact of an earthquake can be reduced using a timely and accurate earthquake early warning system (EEWS) that predicts earthquake parameters. In an onsite EEWS, the warning is sent to areas using the informative primary waveform (P wave) to predict the parameters of the destructive secondary waveform (S wave) of seismic waves [1]. The warning time is usually short to reduce damage and early initiation of the rescue operation. Magnitude estimation is a key component in predicting earthquake impact [2]. Therefore, it is treated as an important subject worth focusing on for the onsite EEW system. Currently, many EEWS worldwide use simple parameter regression relations to predict magnitude [3]. For example, a study shows that recent alarming systems for the Taiwan dataset in EEWS use cumulative absolute absement or peak displacement based regression relations by Lin and Wu (2025) [3]. However, regression relations that use empirically defined terms with partial information limit the performance of

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magnitude estimation [1] and can be replaced by a more advanced framework, such as deep learning. The magnitude estimation is shown to perform better when treated as a timeseries prediction problem rather than a tabular parameter prediction problem [1,4,5]. Thus, the recent trend involves the use of complex artificial intelligence models for timeseries prediction tasks [6]. However, the millions of learnable parameters in these networks require high computational hardware even for inference. The computational problem have been addressed in this study by using a scalable echo state network architecture.

EEWS magnitude prediction has a history of state-of-the-art networks. The deep neural network model proposed by Apriani et al. (2021) [7], which operates at a sampling rate of 80 Hz, is specifically designed for earthquake magnitude estimation using seismic data from the Indonesia region. The deep convolutional network, EEWNet [2] predicts earthquake magnitude (4 to 9) using the first 3 s of the P wave, but is limited to epicenter distances of 25 to 200 km. Unlike convolutional neural network (CNN) [8] based models, the sequence learning models capture temporal dependencies. Bilal et al. (2023) [9] employed a stacked, normalized recurrent neural network (RNN) architecture [10] to forecast earthquakes in the Turkish region. RNNs use high memory and employ error backpropagation (BP) [11]. One major drawback of BP is its sensitivity to bifurcations, which can lead to divergence during training, high computational cost, and suboptimal local minima [12]. Additionally, the training process is often affected by the exploding gradient problem. The long short term memory (LSTM) [10] inverse Correlation framework was proposed by Devi et al. (2024) [13] for earthquake forecasting in the Chile region. However, each LSTM unit contains multiple fully connected layers, making it training computationally intensive for deep networks with long time spans [14]. The large number of parameters also increases the risk of overfitting. Unlike prior EEW magnitude prediction studies [2,9] that rely on linear readouts or end-to-end deep architectures without explicitly addressing data imbalance, this work integrates a dendritic neural network (DENN) [15] readout and a DENN-based CTGAN framework to enhance nonlinear expressivity and improve generalizability across underrepresented magnitude ranges.

Ensembling technique for magnitude prediction by combining CNN and sequence to sequence learning models add generalization. This includes a CNN and an LSTM-based architecture for earthquake magnitude prediction for 0 to 5.7 M_w by Mousavi and Beroza (2019) [4]. Chakraborty et al. (2022) [16] utilized a convolutional network followed by an LSTM network for magnitude prediction. Yoon et al. (2023) [17] utilized a dense CNN and LSTM based model architecture with an embedding network for EEW magnitude prediction. After CNN and LSTM ensemble, the focus shifted to more advanced transformer based networks. Transformers have demonstrated strong capabilities in sequence modeling and scalable design, which has driven a increase of interest in adapting them for time series forecasting. Saad et al. (2022) [5] employed a Vision Transformer network for magnitude prediction using 30 s three component seismograms from Texas seismic data, focusing on events within a narrow magnitude range of 0 to 5.7 M_L. Ge et al. (2024) [18] utilized a Fourier Transformer based architecture for magnitude prediction. However, these networks increase computational cost due to the ensembling of many different units. While prior models [5,18] improve accuracy through architectural depth and attention mechanisms, the proposed work demonstrates that comparable predictive performance can be achieved using a computationally efficient reservoir framework.

The advanced EEWS used in the USA, called ShakeAlert, employs EPIC for rapid point-source estimation and FinDer for finite fault rupture detection using early P wave signals. However, these physics based algorithms rely on predefined models and multi station data, limiting adaptability and accuracy compared to machine learning approaches, and offer scope for improvement [19]. Another model, installed in New Zealand, the Eastern Ruapehu Lahar Warning System [20], uses only threshold based logic to trigger alarms. More advanced existing EEW, such as the CRED framework [21], detect earthquake signals using a 30 s long window for real implementation to continuous data recorded in Guy-Greenbrier, Arkansas, using CNN and LSTM networks.

The most widely adopted approach focuses on component adaptation on timeseries, including Autoformer [22]. These architectures particularly refine the attention mechanism to capture temporal dependencies and to optimize long-sequence complexity. However, the rise of larger lookback windows degraded performance and introduced larger computation explosion as a challenge in these frameworks [6]. Another challenge is that, although the ordering of time steps is crucial in time series, these models often use permutation-invariant attention along the temporal axis, thereby ignoring the sequential order [23]. Together, these factors reduce the Transformer's ability to learn rich representations and model dependencies across variables, ultimately limiting both its effectiveness and generalization on diverse datasets [6]. To overcome the above limitations, iTransformer [6] and Reformer [24] use reversible layers in the Transformer. However, these layers can introduce numerical errors that accumulate across layers, potentially reducing overall model performance. To cut off cost computation in transformer framework, the Flowformer [25] has been introduced. However, the predictive effectiveness is reduced. In this study, earthquake magnitude prediction is treated as a timeseries prediction problem, and representative transformer architectures including Autoformer [22], iTransformer [6], Reformer [24], and Flowformer [25] are used as benchmark models. Unlike these computation intensive frameworks, which may hinder rapid disaster warning in resource constrained regions, the proposed approach achieves both lower computational cost and reduced prediction error. This demonstrates that increasing architectural complexity for generalized timeseries modeling does not necessarily result in superior performance for complex seismic energy pattern prediction tasks.

The high computation requirement problem is addressed recently by Echo State Networks (ESNs) [26] that provide a middle ground. Over the past decade, reservoir computing has gained attention as an alternative to gradient-based approaches for training recurrent neural networks [27]. ESNs have long been leveraged for chaotic atmospheric and hydrological signals, including rainfall runoff forecasting, wind speed prediction, and geomagnetic indices [14]. Therefore, ESNs offer greater learning efficiency for physics-based signals compared to complex architectures [14]. ESN addresses convergence challenges and reduces computational costs by using a least-squares approach for training rather than traditional methods [14]. ϕ-ESN by Gallicchio and Micheli (2011) [28] extends a standard ESN by projecting its reservoir into a larger, randomly weighted space via an Extreme Learning Machine, thereby capturing richer temporal dynamics for improved predictions. However, it relies on a single-scale reservoir. To overcome this, multi-reservoir ESN models have been developed that use multiple reservoirs to improve prediction performance [14].

Research on DeepESN architectures [29] has shown that stacking reservoirs can significantly increase short-term memory capacity and enrich feature representations [30]. Inoue et al. (2024) [31] and Sun et al. (2024) [14] utilized multi-scale ESN to achieve better performance than regular ESN. Consequently, DeepESN [29] and its variants [32] often achieve better prediction performance than standard single layer ESN models [14]. Deep-ESNs are well-suited for handling nonlinear dynamic signals with multi-scale temporal patterns [14]. However, most ESN studies rely on a *single-scale* leak factor that fails to capture the broadband nature of earthquake sources. Leaking rate controls the decay of neuron dynamics, effectively adjusting how much past information influences the current state. By tuning the leaking rate for each layer, Deep-ESNs can achieve layer-specific temporal responses without the rapid signal loss that occurs when scaling input weights [33]. The long short term ESN (LS-ESN) by Zheng et al. (2020) [34] consists of three independent reservoirs, each designed with distinct recurrent dynamics to capture temporal dependencies at different time scales within the time series. Further, advancement involves the Long short-term cross ESN (LSCross-ESN) by Jiang et al. (2025) [35]. It enables the network to capture and fuse multi-scale temporal dependencies by allowing information exchange between reservoirs with delayed and sliding-window connections, preserving both long- and short-term features for better time series prediction. Inter-ESN [36] introduces bidirectional links between two reservoirs, enhancing the exchange of internal states and strengthening the model's overall representational capacity. However, in the case of earthquake signals, where high magnitude depends on long-time phase and low magnitude depends on shorter-time phase, it is complicated to consider both using current ESN frameworks. Multi-leak ESNs can smoothly regulate temporal scales via leak rates, whereas LSCross-ESN [35] relies on discrete reservoirs and delay settings, which are harder to balance [31]. As a result, clear design guidelines for reservoir behavior are still lacking. To avoid the trial-and-error method and the Bayesian optimization approach, we utilized multiscale leaky rate methods that exhibit multiple time-scale dynamics to enhance nonlinear dynamic timeseries prediction. Another important aspect of this study is replacing the linear or traditional readouts with neurophysically inspired readouts. Neurophysiology reveals that biological neurons integrate inputs over multiple membrane time constants and employ nonlinear dendritic sub-compartments to enrich their computational repertoire [37]. Transplanting this principle to artificial networks has triggered a wave of DENN [15] in which each hidden unit is partitioned into sub-compartments that perform quadratic or threshold operations on restricted input subsets. At equal parameter count, dendritic ANNs exhibit higher robustness and sample efficiency than classical multi-layer perceptrons [38]. Poirazi and Mel (2001) [39] demonstrated that dendritic structures can achieve high capacity through structural learning and random synapse formation. The DENN network differs by modeling both the nonlinear response and the localized nonlinearity. Localized nonlinearity refers to a synapse affecting nearby synapses in a nonlinear manner, rather than influencing only the entire neuron [15]. Recently, DENN has been utilized for earthquake magnitude and peak ground acceleration prediction by Joshi et al. (2025) [1], which represents its success in EEW prediction tasks. These advances motivate our adoption of a multiscale leakage vector combined with dendritic readouts for EarthESND.

The work has been extended to address data imbalance by generating a synthetic dataset using conditional tabular adversarial networks (CTGAN) [40], with a dendritic neural network in the generator and discriminator blocks of the GAN. Data augmentation using GANs has been shown to be an effective technique for handling data imbalance in earthquake magnitude prediction [41]. Therefore, in this study, we employ CTGAN to generate synthetic tabular data, providing new insights into data augmentation for this task. The shallow ML techniques are still utilized to handle earthquake magnitude prediction [42]. Thus, the generated synthetic data are integrated into the machine learning (ML) module, where they are used to enhance and average the final magnitude predictions. Artificial intelligence techniques require a well balanced dataset, especially in earthquake magnitude prediction, where the dataset is more heavily weighted toward small events. Recent work by Joshi et al. (2025) [41] proves that onsite magnitude prediction could be improved using data imbalancing techniques. Based on the study by Joshi et al. (2025) [41], we applied CTGAN [40], enhanced with dendritic neural networks in both the generator and discriminator, are used to generate synthetic tabular data samples.

Translating these principles to ESNs suggests (i) using a *vector* of leak rates to maintain parallel memories spanning 10^{-1}–10^{1} s, and (ii) replacing the plain affine read-out with dendrite-like quadratic mixing to boost expressiveness at negligible training cost. Biological neurons are more complex and superlinear in nature [15] than typical neural networks [43]. The presence of active channels, along with the leaky properties of dendritic membranes, indicates that a single synaptic input may exert a nonlinear effect on nearby synapses [15]. This study introduces **EarthESND**, a *multi-scale Echo State Network with dendritic read-outs* tailored for real-time magnitude prediction of events recorded in Japan using 2, 3, 4, 5, and 6 s of P-wave dataset. The key contributions of the proposed work are: (1) *Cost Effective modeling:* Unlike prior EEW magnitude prediction approaches that rely on computationally intensive deep learning models, this study emphasizes the use of reservoir computing on low-cost devices, achieving comparable prediction performance with reduced computational demands. (2) *Multiscale reservoir:* This work employs a multiscale, deep ESN architecture for efficient time series prediction. A log-uniform leak vector enables simultaneous tracking of fast P-wave onsets and slower cumulative-energy build-up. (3) *Dendritic read-out:* The final layer is replaced with a more expressive dendritic neural network, which also incorporates physics-based earthquake parameters as additional tabular inputs, making the framework novel compared to existing EEW prediction models. (4) *Synthetic Data Generation:* The study is further extended to address data imbalance by generating synthetic samples using a dendritic neural network based generator and discriminator units in a CTGAN framework, namely, dendritic conditional tabular generative adversarial network (DENN-CTGAN). (5) *Extensive Result Study:* Experiments were conducted on recent efficient time-series transformers, novel ESN frameworks, and state-of-the-art regression and deep learning models for magnitude prediction, followed by a detailed analysis of the 2024 Noto earthquake and inter-region Indian earthquake analysis using the proposed framework.

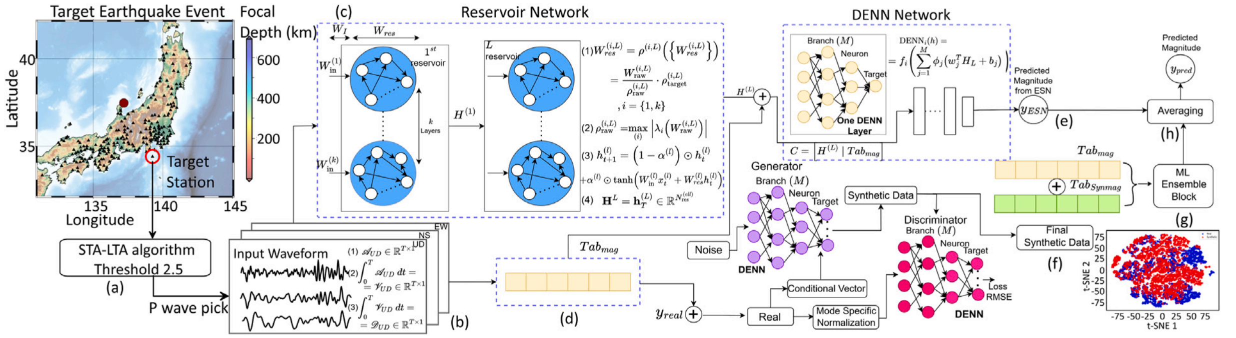

Fig. 1. End-to-end EarthESND workflow for onsite EEW magnitude estimation. (a) The triggered strong-motion seismogram is recorded using the short-term average and long-term average algorithm [44]. (b) The processed waveforms are passed to the (c) reservoir network. (d) Extraction of physics-based tabular features from input waveforms. (e) The concatenated input features from the reservoir and tabular features are passed to DENN for magnitude prediction, y_{ESN}. (f) The synthetic dataset is generated using DENN-CTGAN. (g) The synthetic and real datasets are passed to the ML ensemble block. (h) The final result, y_{pred}, is the average of the predicted magnitude from the ML Ensemble block and y_{ESN}.

Motivation. The utilization of low-end computational machines for earthquake parameter prediction enables the wide resolution to develop cost-effective yet efficient EEW systems. This approach achieves comparable performance with minimal computational cost, enabling wider participation in research and the practical deployment of EEW.

2. Methodology

2.1. Problem formulation and notation

The raw accelerograms of three components, named as north-south (NS), east–west (EW), and up-down (UD), of early N seconds of P wave are passed as input to the model as shown in Fig. 1(a). These raw acceleration waveforms are transformed into processed waveform $\mathcal{A}_c \in \mathbb{R}^{T\times 1}, c \in \{NS, EW, UD\}$ using baseline correction, station specific scaling, and transformed into velocity, $\mathcal{V}_c \in \mathbb{R}^{T\times 1}, c \in \{NS, EW, UD\}$, and displacement, $\mathcal{D}_c \in \mathbb{R}^{T\times 1}, c \in \{NS, EW, UD\}$. $T = S \times N$ denotes the sampling frequency times seconds (100 Hz for the target dataset).

$\mathbf{X}_i = \{\mathcal{A}_{NS}, \mathcal{A}_{EW}, \mathcal{A}_{UD}, \mathcal{V}_{NS}, \mathcal{V}_{EW}, \mathcal{V}_{UD}, \mathcal{D}_{NS}, \mathcal{D}_{EW}, \mathcal{D}_{UD}\} \in \mathbb{R}^{T\times C}$ is a multichannel seismograms of length T sampled. Channel size $C = 9$, corresponding to the concatenation of three-component acceleration $\mathbf{a}(t)$ and the co-integrated velocity $\mathbf{v}(t) = \int \mathbf{a}(t)\,dt$ and displacement $\mathbf{d}(t) = \int \mathbf{v}(t)\,dt$ as shown in Fig. 1(a). Our goal is to learn a mapping $f_\theta : \mathbb{R}^{T\times C} \to \mathbb{R}$ such that $\hat{M}_i = f_\theta(\mathbf{X}_i)$. Some prominent tabular features ($Tab_{mag} \in \mathbb{R}^{6\times 1}$) that correlate well with earthquake magnitude are predicted from the waveform. The waveform $\mathcal{X}_i \in \mathbb{R}^{T\times 9}$ for ith record and $Tab_{mag} \in \mathbb{R}^{6\times 1}$ are passed as input to the proposed architecture. The current multi-scale ESN with dendritic readouts produces deterministic point estimates of magnitude (continuous real values). The magnitude (y_{pred}) in M_{JMA} scale is predicted as output.

P phase selection. Prior to modeling, we reject traces that do not contain an event onset through a two-stage gate. First, the earthquake P wave recording from the station is picked using the short-term average and long-term average algorithm (STA/LTA) pick [44] with a threshold of 2.5. This threshold was empirically selected to reliably capture the onset of $M_{JMA} \geq 3$ strong-motion seismogram, ensuring early and stable detection of the initial P-wave arrival. Then apply the scaling factor corresponding to each station. To further suppress small-amplitude fluctuations and spurious triggers, we compute the mean of the first 20 samples of each candidate window; only windows with a mean acceleration exceeding 0.01 gal are passed to the ESN. After STA–LTA detection, we perform baseline correction to remove baseline drift by subtracting the mean of the first 20 samples [45]. Then apply Butterworth bandpass filtering with a 4-pole bandpass filter of 0.0075 to 45 Hz to clean the waveform and remove the original shift. Consequently, the present framework is intended for event-containing earthquake records, and performance on arbitrary continuous streams or extremely low-SNR traces outside this gate is not applicable. Traces with missing waveform segments or weak signals are automatically rejected during preprocessing using a mean amplitude criterion of 0.01 gal computed over the first 20 samples. This ensures that only clear, strong motion signals containing P-wave records are passed to the model. Previous studies have shown that longer P wave durations lead to more reliable magnitude estimation, but the lead time cost decrease [46]. Thus, in practice, researchers only need to pass a strong-motion seismogram into the model, and if the event magnitude is greater than M_{JMA} 3, the model will automatically detect and estimate it without any manual intervention.

Reservoir dynamics. An Echo State Network projects the sequence $\mathbf{X}_i$ into a high-dimensional state trajectory as given by $\mathbf{h}_{t+1} = (1-\boldsymbol{\alpha}) \odot \mathbf{h}_t + \boldsymbol{\alpha} \odot \tanh\big(\mathbf{W}_{in}\mathbf{x}_t + \mathbf{W}_{res}\mathbf{h}_t\big)$, where $\mathbf{h}_t \in \mathbb{R}^{N_{res}}$, $\mathbf{x}_t \in \mathbb{R}^C$ is the input at time t, $\mathbf{W}_{in} \in \mathbb{R}^{N_{res}\times C}$ and $\mathbf{W}_{res} \in \mathbb{R}^{N_{res}\times N_{res}}$ are fixed random matrices, and $\boldsymbol{\alpha} \in (0,1]^{N_{res}}$ is the vector of leak rates. We retain only the *terminal* state $\mathbf{h}_T$ as a sequence fingerprint; thus the design matrix is $\mathbf{H} = [\mathbf{h}_1^\top, \ldots, \mathbf{h}_N^\top]^\top \in \mathbb{R}^{N\times N_{res}}$. The ESN formulation preserves causality and offers $\mathcal{O}(N_{res}C)$ compute per sample, orders of magnitude lighter than back-propagated RNNs [26]. A dendritic neural network read-out then yields the prediction.

Dendritic neural network. The first step in DENN is a composition of dendritic layers that defines a function $DENN(x) = \{f_{\text{out}} \circ f_L^D \circ \cdots \circ f_1^D(x)\}$, where $f_l^D : \mathbb{R}^{n_{l-1}} \to \mathbb{R}^{n_l}$ represents the dendritic layer l [15]. Each layer contains n_l neurons, and each neuron consists of d dendritic branches. Each branch receives a subset of inputs (n_{l-1}/d) from the previous layer and performs a local weighted summation followed by a nonlinear transformation: $h_{l,i,j}(x_{l-1}) = \phi_j\left(\sum_{m=1}^{n_{l-1}}(S_{l,i,j,m} W_{l,i,j,m}) x_{l-1,m} + b_{l,i,j}\right), \quad j \in [d]$, where $S_{l,i,j,m} \in \{0,1\}$ denotes whether the mth input connects to the jth branch of the ith neuron, $W_{l,i,j,m}$ are the synaptic weights, and $\phi_j(\cdot)$ is the local dendritic nonlinearity (e.g., sigmoid or Gaussian). The neuron output aggregates all dendritic branch responses: $g_{l,i}(x_{l-1}) = f_i\left(\sum_{j=1}^{d} h_{l,i,j}(x_{l-1})\right) + b_{l,i}$, where $f_i(\cdot)$ is the neuron-level activation function and $b_{l,i}$ is the neuron bias. Thus, the dendritic layer output is $f_l^D(x_{l-1}) = [g_{l,1}(x_{l-1}), \ldots, g_{l,n_l}(x_{l-1})]^\top$.. In particular, when $d = 1$ and $\phi(\cdot)$ is linear, the dendritic layer reduces to a standard fully connected layer. The DENN architecture ensures nonlinearity and localized learning by controlling a subset of input passed [15].

2.2. Proposed framework

Deep Echo state Network (DESN) [47] involves incorporating multiple reservoirs and allowing for dense connections for complex representations. Following the neurophysiological observation that cortical pyramidal Cells integrate inputs over multiple membrane time scales [37], we draw a *vector* $\boldsymbol{\alpha} = \left[\alpha^{(1)}, \ldots, \alpha^{(N_{res})}\right]$ from a log–uniform prior $\alpha^{(k)} \sim \mathcal{U}\left(10^{-1}, 1\right)$. The resulting update rule $\mathbf{h}_{t+1} = (1-\boldsymbol{\alpha}) \odot \mathbf{h}_t + \boldsymbol{\alpha} \odot \tanh(\mathbf{W}_{in}\mathbf{x}_t + \mathbf{W}_{res}\mathbf{h}_t)$ creates *parallel fading memories* ranging from 0.1 to 1 s. This design shows that Deep-ESN performance improves by 15–25% on chaotic signals, with empirical support [31].

We stack L multi–scale reservoirs in series and propagate the seismic tensor only *forward* as shown in Fig. 1(c). The output of layer ℓ is fed as the sole input to layer $\ell + 1$. Intermediate states are not concatenated or skip–connected. Formally, the update rule with $\mathbf{h}_t^{(\ell)} \in \mathbb{R}^{N_{\text{res}}^{(\ell)}}$ be the hidden state of layer ℓ at time t is as follows:

$$\mathbf{h}_{t+1}^{(\ell)} = (1-\boldsymbol{\alpha}^{(\ell)}) \odot \mathbf{h}_t^{(\ell)} + \boldsymbol{\alpha}^{(\ell)} \odot \tanh\left(\mathbf{W}_{\text{in}}^{(\ell)}\mathbf{x}_t^{(\ell)} + \mathbf{W}_{\text{res}}^{(\ell)}\mathbf{h}_t^{(\ell)}\right) \tag{1}$$

where $\mathbf{W}_{\text{in}}^{(\ell)} \in \mathbb{R}^{N_{\text{res}}^{(\ell)} \times C^{(\ell)}}$, $\mathbf{W}_{\text{res}}^{(\ell)} \in \mathbb{R}^{N_{\text{res}}^{(\ell)} \times N_{\text{res}}^{(\ell)}}$, and the leak vector $\boldsymbol{\alpha}^{(\ell)} \in (0,1]^{N_{\text{res}}^{(\ell)}}$ are fixed. After the final reservoir ($\ell = L$) we retain only its terminal state $\mathbf{h}_T^{(L)}$. Only the *last* reservoir contributes to the read-out, and gradients are never propagated through any other ESN layer. $\boldsymbol{\alpha}^{(\ell)}$ is the per-neuron leak rate vector. It controls how much of the new input (from $\tanh(\cdot)$) influences the next hidden state versus how much of the previous state is retained [14].

To initialize the reservoir in a multi layer and multi ESN, a random recurrent matrix $W_{\text{raw}}^{(i,\ell)} \in \mathbb{R}^{D^{(\ell)} \times D^{(\ell)}}$ is first generated for ith stack of ℓ^{th} layer reservoir. Each reservoir in each layer follows Eq. (1). $\nu \in (0,1]$ is the sparsity denoting fraction of non-zero weights. Binary masking with $\nu \in (0,1]$ check is done for $W_{\text{raw}}^{(i,\ell)}$ and filled with standard normal distribution ($\mathcal{N}(0,1)$). This improves memory separation and computational efficiency. The spectral radius of this matrix is then computed as $\rho_{\text{raw}}^{(\ell)} = \max_i |\lambda_i(W_{\text{raw}}^{(i,\ell)})|$, where λ_i are the eigenvalues of $W_{\text{raw}}^{(i,\ell)}$. To ensure the desired dynamic behavior of the reservoir, the matrix is rescaled so that its spectral radius matches a predefined target value $\rho_{\text{target}}^{(\ell)}$. This is done using the transformation, for $i^{th} \in \{0, k\}$ layer of reservoir and ℓ^{th} reservoir with k total layers:

$$W_{res}^{(i,\ell)} = \rho^{(\ell)}(W_{res}^{(i,\ell)}) = \frac{W_{\text{raw}}^{(i,\ell)}}{\rho_{\text{raw}}^{(\ell)}} \cdot \rho_{\text{target}}^{(\ell)} \tag{2}$$

This procedure ensures that the internal dynamics of the reservoir are stable and satisfy the echo state property. W_{in} is the input scaling parameter and is initialized with a normal distribution. The output from one reservoir is passed as input to the other reservoir. After processing all T timesteps, the output of the Lth layer is its terminal hidden state: $\mathbf{H}^L = \mathbf{h}_T^{(L)} \in \mathbb{R}^{N_{\text{res}}^{(\ell)}}$.

Data fusion enables more complete and precise insights to support decision-making in disaster management [48]. The earthquake parameters are correlated with physics-based earthquake features derived from the P-waveform. Thus, to decrease prediction error, the output received from the last ESN layer has been concatenated with the tabular feature set. The tabular feature set is denoted by Tab_{mag} as shown in Fig. 1(d). These parameters have been widely used as tabular input in several studies for magnitude prediction [41,49,50] and are described in *Supplementary Section 1.1, Feature Description*. The tabular set consists of characteristic period (τ_c), integrated squared displacement ($ID2$), integrated squared velocity ($IV2$), P wave index (PI), root sum of squared velocity ($RSSCV$), peak velocity acceleration ratio (T_{va}), and cumulative absolute velocity (CAV). These parameters are obtained from the vertical component of the P waveform since the signal-to-noise ratio is better in the vertical component than in the horizontal components. This makes them more reliable for high-quality earthquake magnitude analysis [51].

Linear read-outs are the norm in reservoir computing because they admit a closed-form solution, yet they implicitly assume that the reservoir state is linearly separable with respect to the target. We augment the read-out with a DENN to raise the expressive ceiling without iterative back-propagation. The final reservoir's output is concatenated with tabular features, $C = [H^{(L)}|Tab_{mag}]$. It is then passed through $DENN(C(t);\theta)$. We obtain the magnitude prediction by a K layer dendritic neural: $y_{pred} = DENN_{i+1}...(DENN_1(C(t);\theta_1),\theta_{i+1}), i \in \{0,\ldots,K\}$. There are two DENN layers utilized in this network that give optimal performance. Stochastic regularizers like dropout and activation functions like rectified linear unit remain distinct [52], even though both influence a neuron's output. The Gaussian error linear units (GELU) activation is utilized to provide a smooth nonlinearity that can be interpreted as the expected value of an input-dependent stochastic dropout-like mask. The GELUs [52] activation have been utilized in $DENN_1$ and is represented as: $\text{GELU}(x) = \frac{1}{2}x\left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}(x + 0.044715\,x^3)\right)\right)$. The linear activation has been utilized in the $DENN_2$ layer. The two DENN layers, $DENN_1$ and $DENN_2$ are used after reservoir network units (n_l) of 64 and 1 for all five datasets. The branches (d) of $DENN_1$ are 2, 2, 3, 3, and 2 are 2SE, 3SE, 4SE, 5SE, and 6SE datasets. The sparsity ($\nu \in (0,1]$) of the 2SE, 3SE, 4SE, 5SE, and 6SE datasets is 0.1, 0.1, 0.2, 0.1, and 0.1. The prediction from the final DENN layer is the predicted magnitude using seismograms passed to the reservoir network and is denoted by y_{ESN} as shown in Fig. 1(e). The adaptive moment estimation (Adam) optimizer is utilized in this work, which combines the advantages of handling sparse gradients (increases the learning rate for rarely-updated parameters) and handling non-stationary settings (mini-batches that vary a lot) [53]. Moreover, unlike stochastic gradient descent, each parameter get its own adaptive weights [53]. The learning rate of the reservoir network for all five datasets is set to 0.001, 0.0014, 0.001, 0.001, and 0.001. The mean squared error (MSE) is used as the loss function. The multiple scales are obtained from multiple leaky rates, which are selected as 0.8, 1.0, 0.5, 0.1, and 0.08 for all dataset (2SE to 6SE). The epoch size is 50 for all five datasets, and batch_size is 512 for all datasets.

2.3. Generating synthetic dataset

The imbalance causes the model performance to be skewed towards the majority magnitude cases. The dendritic neural network-based CTGAN delivered in the proposed work acts as an answer to this problem. CTGAN employs mode-specific normalization to effectively represent multimodal and non-Gaussian data distributions of continuous columns. The mode-specific normalization utilizes a Variational Gaussian Mixture model [54] to approximate its distribution of continuous column (C_i). The learned Gaussian mixture for column C_i is expressed as: $P_{C_i}(c_{i,j}) = \sum_{p=1}^{m_k} \mu_p \mathcal{N}(c_{i,j}; \eta_p, \theta_p)$, where μ_p and θ_p represent the weight and standard deviation of the pth mode, respectively. The conditional generator in CTGAN aims to reproduce the true data distribution, which can be formulated as: $\mathbb{P}(\text{row}) = \sum_{k \in D_{i^*}} \mathbb{P}_G(\text{row} \mid D_{i^*} = k^*) \mathbb{P}(D_{i^*} = k)$ [40]. The DENN network serves as both the generator and discriminator in the CTGAN architecture, referred to as DENN-CTGAN, which is employed in this study for synthetic data generation, as shown in Fig. 1(g). The t-Distributed Stochastic Neighbor Embedding (t-SNE) [55] elevates the crowding and the optimization problems of SNE and are used in our study to visualize actual and synthetic high dimensional data as shown by Fig. 1(g) for 3SE dataset. t-SNE Plots comparing synthetic and real input output parameters generated by DENN-CTGAN are presented in *Supplementary, Figure 1*. A total of 10000 synthetic records are generated using the features Tab_{mag} and target values y_{act} from the training set, which serves as inputs to the DENN-CTGAN model. The resulting synthetic dataset is denoted as $\{Tab_{syn} \mid y_{syn}\}$. The augmented dataset is the combination of $\{Tab_{aug}|y_{aug}\} = \{Tab_{syn}|y_{syn}\} \parallel \{Tab_{act}|y_{act}\}$. These records are sent to the ML Ensemble block for training. The ML ensemble block consists of the XGBoost (XGB) [56], Light GBM (LGBM) [57], and CatB [58] learners trained using an augmented dataset and a real dataset. Each learner is trained twice: once using the augmented dataset $\{Tab_{aug} \mid y_{aug}\}$ and once using the real dataset $\{Tab_{act} \mid y_{act}\}$. Thus, six ensemble models are obtained in total. The final magnitude prediction is computed by averaging the outputs of these six ML models with the y_{ESN} result from the reservoir framework, as illustrated in Fig. 1(h).

3. Dataset

Selected area. The seismicity rate in Japan is the highest, and large earthquakes are frequently experienced in Japan. These earthquakes are due to the complex geodynamic context of oceanic plate interaction. These plates include the Philippine and Pacific sea plates that subduct beneath the Eurasian plate [49]. These plate interactions cause deep, intermediate, and crustal earthquakes in Japan. The raw strong-motion seismograms of the earthquake dataset have been organized in the Kyoshin Network (K-NET) repository, maintained by the Japan Meteorological Agency (JMA) catalog [59]. The database's accuracy and the variety of events it contains make it a crucial dataset for understanding seismic signatures.

Selected events. Fig. 2 represents the (a) station and (b) epicenter location of the selected earthquakes. Events are split into 70:15:15 of train, test, and validation datasets with 25337, 5429, and 5430 records from 1996 to July 2024 as shown in Table 1. The Fig. 2(c) represents the magnitude distribution of the test, train, and validation sets with respect to epicenter distance. The qualitative overall counts of the train, test, and validation splits are shown in Fig. 2(d) and (e) for earthquake epicenter distance and magnitude. Fig. 2(f) and (g) represent the scatter plot of focal depth and epicenter distance, and the count of focal depth with respect to the dataset. We used the train:test:validation split with well-established practical heuristics commonly used in applied machine learning, and validated these choices empirically via quantitative analysis of random splits [60]. Care was taken to address the imbalance in magnitude distribution in real events. Since the magnitudes are continuous, they were divided into discrete bins ranging from 3.0 to 8.0 with an interval of 0.5 using np.arange(3.0, 8.5, 0.5) to ensure stratified sampling across the magnitude range. We have separated the Noto event 232 records for testing. *Supplementary Table 2* shows that the 70:15:15 ratio was selected as the optimal ratio set for the proposed prediction problem. In this work, the dataset is segmented into windows of 2, 3, 4, 5, and 6 s from the initial portion of the P-wave for magnitude prediction. The training, testing, and validation records remain the same across all segments, with only the time window varying. These datasets are referred to as 2SE, 3SE, 4SE, 5SE, and 6SE, corresponding to the 2, 3, 4, 5, and 6-second P-wave windows, respectively. The developed model imposes no lower or upper limits on the selection of epicenter distance, focal depth, or hypocenter distance. Only earthquakes with magnitudes ranging from 3.0 to 7.6 M_{JMA} were considered, thereby reducing the influence of outliers to prevent the high data imbalance problem. Another reason for selecting magnitudes below 8 M_{JMA} is that the analysis by Noda and Ellsworth (2016) [61] shows that for earthquakes of magnitudes 5, 6, and 7 M_w, the departure period emerges approximately 0.38 to 1.5 s, 1.05 to 4 s, and 2.87 to 10 s after the P-wave arrival. Thus, the 2- to 6-second P-wave window can rarely contain the departure time information for greater than 7 M_{JMA}.

We have used earthquake events recorded in the Indian strong-motion seismograms to assess the generalizability of the framework in areas of sparse coverage, as shown in Fig. 2(h). These events were taken from the "Program for Excellence in Strong Motion Studies (PESMOS)" database, curated by the Indian Institute of Technology, Roorkee, India (www.pesmos.in). These events form the inter-regional earthquake dataset consists of 157 records of 12 earthquakes. Details of this dataset are provided in Table 2.

Table 1
The maximum and minimum values of parameters used in the training, testing, and validation datasets.

Set	Records	Parameter							
		Magnitude (M_{JMA})		Epicenter Distance (km)		Focal Depth (km)		Hypocenter Distance (km)	
		Min	Max	Min	Max	Min	Max	Min	Max
Training	25 337	3.0	7.7	1.08	1607.85	0	619	5.57	1668.29
Testing	5 429	3.0	7.7	2.87	1253.71	0	619	9.49	1341.38
Validation	5 430	3.0	7.7	2.02	1404.37	0	619	7.30	1483.17

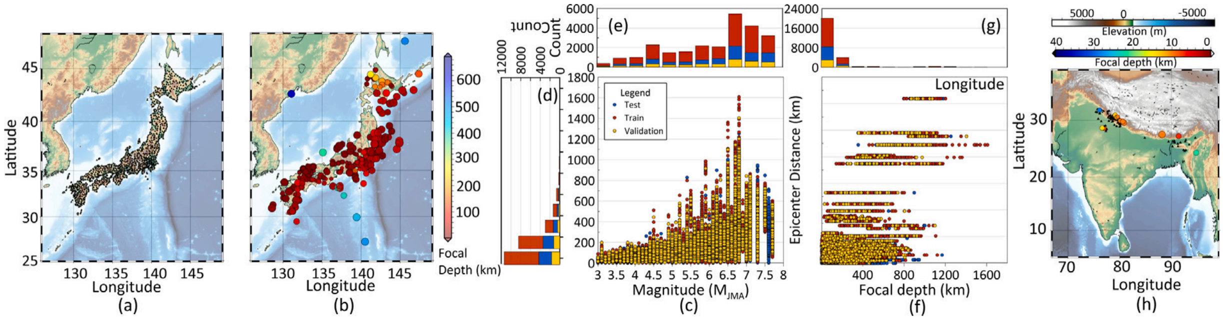

Fig. 2. Geographical visualization of the: (a) station and (b) earthquake epicenters. The color bar indicates the focal depth (km). (c) Scatter plot of magnitude against epicentral distance (km). (d) Distribution of epicentral distances and (e) magnitude distributions for the training, validation, and test sets. (f) Focal depth and epicentral distance. (g) Focal depth distribution across data splits. (h) The spatial distribution of the Indian region stations and earthquake epicenters with 157 records. The color bar represents focal depth (km) and land elevation (m).

Table 2
Dataset of inter-region earthquakes occurring in India, Myanmar, Bhutan, and Nepal with 157 records and 12 earthquakes.

Region	Date	No. of Stations	Magnitude (M_w)	Focal Depth
Myanmar-India Border	11-08-2011	12	5.6	12
Uttarkashi, India	21-09-2009	10	4.7	12
Bhutan	31-12-2009	5	5.5	5
Punjab - Himachal Border, India	14-03-2010	12	4.6	13
Bageshwar, Uttarakhand, India	01-05-2010	6	4.6	13
India- Nepal Border	04-04-2011	23	5.7	24
Nepal	01-12-2016	20	5.2	20
Chamoli, Uttarakhand, India	20-06-2011	11	4.6	7
Sonipat, India	07-09-2011	6	4.2	7
Sikkim, India -Nepal Border	18-09-2011	13	6.8	9
Bahadurgarh, India	05-03-2012	19	4.9	21
Chamoli, India	29-11-2015	10	4.0	14

4. Result

Metrics includes percentage mean absolute error (MAE), root mean squared error (RMSE), and improvement, and a down arrow ↓ is used . The percentage improvement of our model over another model in terms of MAE is given by: Percent Improvement = $\frac{\text{MAE}_{\text{other}} - \text{MAE}_{\text{ours}}}{\text{MAE}_{\text{other}}} \times 100\%$. Down (↓) arrows are included next to each performance metric to indicate that lower values are better.

4.1. Ablation study and parameter analysis

In this paper, ablation study and parameter analysis refer to systematically removing or modifying model units and changing hyperparameters. Selecting the right parameters is essential for building an optimal ESN. The layer configuration and corresponding units yielding the lowest error are discussed in detail in *Supplementary Section 2.1* as the optimal setup for the proposed model. Table 3 presents the ablation study performed by replacing and adding different units to identify the best-performing configuration. The improvement in model performance based on MAE is 8.0%↓, 9.4%↓, 8.1%↓, 6.8%↓, and 8.8%↓ compared to the model when a fully connected neural network (FCNN) is replaced with DENN for the 2 to 6SE dataset. The ablation study shows that the ESN network with DENN performs better than the FCNN-based reservoir readout. Thus, justifying the selection of DENN in the EarthESND framework. The DENN's efficacy is further demonstrated by replacing it with ridge or linear regression. For the 2SE dataset, the error for ridge regression reservoir readout is approximately 22.5%↓, 23.9%↓, 22.7%↓, 23.6%↓, and 23.6%↓ higher than the EarthESND model for 2 to 6SE dataset. While using linear regression as the output layer results in an error about 21.6%↓, 24.7%↓, 22.7%↓,

Table 3
Ablation study to evaluate the individual contribution of each architectural component to the overall system performance on test data of Japan.

Method	2SE		3SE		4SE		5SE		6SE	
	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)
Single ESN Layer DENN	0.81	0.96	0.80	0.95	0.80	0.97	0.79	0.95	0.77	0.91
Single ESN Layer FCNN	0.84	0.99	0.84	0.91	0.82	0.90	0.81	0.91	0.78	0.89
Multiscale Single Layer reservoir + DENN	0.87	1.16	0.86	1.15	0.86	1.14	0.84	1.09	0.82	1.08
Multiscale Single Layer reservoir + FCNN	0.88	1.13	0.88	1.10	0.86	1.04	0.83	0.98	0.81	0.94
Deep ESN (2 stack) + FCNN	1.21	1.53	1.19	1.50	1.18	1.45	1.15	1.40	1.12	1.38
Deep ESN (3 stack) + FCNN	0.91	1.24	0.90	1.21	0.91	1.23	0.90	1.19	0.90	1.21
Deep ESN (4 stack) + FCNN	0.97	1.28	0.95	1.22	0.94	1.20	0.92	1.12	0.90	1.11
Deep ESN (8 stack) + FCNN	0.92	1.02	0.90	1.01	0.89	0.99	0.90	0.98	0.97	1.01
Deep ESN (2 stack) + DENN	1.02	1.42	1.02	1.40	1.00	1.38	0.99	1.35	0.99	1.32
Deep ESN (3 stack) + DENN	0.87	0.99	0.87	0.98	0.87	0.99	0.88	1.00	0.89	1.03
Deep ESN (4 stack) + DENN	0.96	1.25	0.94	1.20	0.94	1.12	0.92	1.08	0.91	1.05
Deep ESN (8 stack) + DENN	0.92	1.12	0.93	1.10	0.92	1.08	0.91	1.05	0.92	0.99
EarthESND with FCNN	0.75	0.98	0.74	0.97	0.74	0.98	0.73	0.96	0.74	0.92
Ridge Regression at output layer	0.89	1.12	0.88	1.11	0.88	1.10	0.89	1.14	0.89	1.14
Linear regression at output layer	0.88	1.11	0.89	1.13	0.88	1.12	0.87	1.08	0.87	1.06
Removed Tabular Input	0.90	1.25	0.91	1.26	0.89	1.21	0.89	1.22	0.88	1.19
RNN network with DENN	0.92	1.22	0.92	1.23	0.91	1.21	0.89	1.19	0.89	1.20
RNN network with FCNN	0.91	1.27	0.91	1.26	0.92	1.27	0.90	1.18	0.90	1.21
LSTM network with DENN	0.88	1.16	0.90	1.21	0.90	1.19	0.88	1.18	0.87	1.14
LSTM network with FCNN	0.92	1.21	0.91	1..22	0.90	1.19	0.90	1.21	0.89	1.18
Only $\mathbf{X}_i = \{\mathcal{A}_{NS}, \mathcal{A}_{EW}, \mathcal{A}_{UD}\}$	0.85	1.14	0.85	1.12	0.85	1.14	0.84	1.12	0.83	1.12
Only $\mathbf{X}_i = \{\mathcal{V}_{NS}, \mathcal{V}_{EW}, \mathcal{V}_{UD}\}$	0.83	1.19	0.84	1.21	0.84	1.20	0.85	1.23	0.82	1.16
Only $\mathbf{X}_i = \{\mathcal{D}_{NS}, \mathcal{D}_{EW}, \mathcal{D}_{UD}\}$	0.80	1.09	0.82	1.13	0.80	1.10	0.80	1.11	0.82	1.16
CTGAN-NN	0.78	0.97	0.79	0.99	0.79	0.98	0.77	0.96	0.77	0.97
CTGAN-Conv	0.83	1.25	0.83	1.21	0.82	1.19	0.82	1.18	0.81	1.17
ML Block: RF and XGB	0.78	0.96	0.78	0.97	0.77	0.96	0.77	0.95	0.76	0.94
ML Block: XGB and LGBM	0.76	0.94	0.76	0.93	0.75	0.94	0.74	0.93	0.74	0.92
ML Block: CatB, RF, and XGB	0.72	0.92	0.72	0.93	0.71	0.92	0.71	0.92	0.72	0.93
Only ML Block ($\{Tab_{aug} \vert y_{aug}\}$)	0.98	1.28	0.97	1.26	0.97	1.27	0.96	1.26	0.96	1.27
Only ML Block ($\{Tab_{syn} \vert y_{syn}\}$)	1.05	1.34	1.05	1.35	1.01	1.30	1.02	1.31	0.98	1.28
Only ML Block ($\{Tab_{act} \vert y_{act}\}$)	1.03	1.37	1.03	1.35	0.99	1.30	0.98	1.30	0.96	1.25
Vanilla ESN	0.89	1.15	0.88	1.10	0.86	1.09	0.86	1.02	0.86	1.01
Deep ESN + Linear	0.84	1.03	0.83	0.98	0.83	0.98	0.81	0.97	0.81	0.97
Deep ESN + DENN	0.78	0.96	0.76	0.97	0.76	0.96	0.75	0.97	0.75	0.97
Deep Multiscale ESN+ DENN	0.74	0.96	0.73	0.96	0.73	0.95	0.74	0.96	0.72	0.95
Deep ESN + DENN+ Multiscale reservoir+ ML Block + DENN-CTGAN (Proposed)	**0.69**	**0.86**	**0.67**	**0.84**	**0.68**	**0.86**	**0.68**	**0.85**	**0.68**	**0.85**

21.8%↓, and 21.8%↓ higher than proposed network for 2 to 6SE dataset. Thus, demonstrating the efficacy of the DENN model over regression-based reservoir readout, as shown in Table 3. Similarly, the EarthESND model is compared with sequence-to-sequence learning models using simple RNNs and LSTMs. As shown in Table 3, the ESN-based model with layering and scaling outperforms the simpler RNN and LSTM approaches.

Fig. 3 represents the actual and predicted magnitude using early (a) 2, (b) 3, (c) 4, (d) 5, and (e) 6 s of P waveform. Fig. 3 represents the error (predicted-actual) obtained with respect to epicenter distance for (f) 2, (g) 3, (h) 4, (i) 5, and (j) 6 s of P waveform. It can be visualized from Fig. 3 (f-j) that the nearby epicenter distance gives more error due to the collapsing of the P waveform on the S waveform. Therefore, more energy information is released in the P waveform [1]. The error increases with increasing epicentral distance because seismic waves lose more energy at greater epicentral distances. This energy reduction reduces the clarity of the waveform features related to the earthquake source [41]. Thus, the epicenter distance affects the model's magnitude-prediction error. One major limitation of EEW parameter prediction is the data skewness toward moderate to low magnitude events (less than 5 M_{JMA}).

4.2. State-of-the-art comparison

In this study, all the previous state-of-the-art deep learning frameworks for timeseries were retrained using the same training, testing, and validation datasets as the proposed model to ensure a fair and consistent comparison. Thus, the predicted magnitude is in the M_{JMA} scale. Table 4 represents the quantitative analysis of the MAE and RMSE for comparison of EarthESND performance with other state-of-the-art methods. All models were tuned using the same protocol, with mean squared error (MSE) as the validation metric and a fixed training budget of 50 epochs. The final hyperparameters are listed in *Supplementary Table 12 and 13.* It can be seen from Table 4 that the EarthESND model outperforms other state-of-the-art methods in terms of MAE and RMSE for 3SE to 6SE dataset. However, for 2SE and 3SE, the DFTQuake [1] performs better then the proposed method. However, the training time of

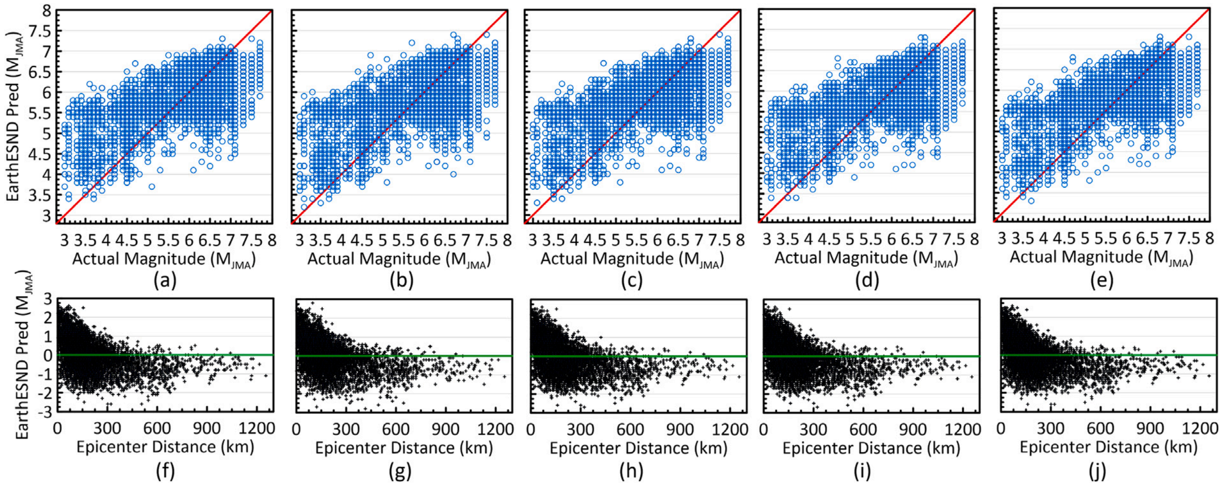

Fig. 3. Actual and predicted magnitudes from the EarthESND model using (a) 2, (b) 3, (c) 4, (d) 5, and (e) 6 s of P-wave data. The red line indicates the ideal 1:1 relation. Prediction errors (predicted - actual) versus epicentral distance (km) for (f) 2, (g) 3, (h) 4, (i) 5, and (j) 6 s inputs are also shown; the green line represents zero error.

Table 4
Comparison of state-of-the-art ESN frameworks, deep learning methods for early earthquake magnitude prediction, and recent time series forecasting transformer frameworks on test data of Japan. The boldface represents the least error. Metrics are in M_{JMA}.

Method	2SE		3SE		4SE		5SE		6SE	
	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)
LSCross-ESN [35]	0.99	1.15	0.99	1.15	0.99	1.15	0.98	1.12	0.96	1.10
LS-ESN [34]	1.01	1.21	0.99	1.18	0.98	1.02	0.97	1.11	0.97	1.08
Inter-ESN [36]	0.96	1.21	0.95	1.20	0.94	1.18	0.94	1.15	0.92	1.12
ϕ-ESN [28]	0.96	1.14	0.95	1.11	0.94	1.12	0.93	1.11	0.92	1.08
Vanilla ESN [26]	1.21	1.15	1.08	1.10	1.08	1.08	1.02	1.12	1.02	1.02
Deep ESN [29]	0.84	1.03	0.83	0.98	0.83	0.98	0.81	0.97	0.81	0.97
Ensemble ESN [32]	1.08	1.39	1.04	1.32	1.01	1.28	0.99	1.25	0.99	1.20
edRVFl [65]	0.94	1.08	0.95	1.06	0.91	1.05	0.90	1.03	0.91	1.02
edESN [66]	1.11	1.42	1.02	1.31	0.99	1.21	0.97	1.17	0.98	1.13
SNRNN [9]	1.12	1.27	1.19	1.34	1.13	1.28	1.02	1.18	1.01	1.04
ViT magnitude [5]	0.70	0.86	0.71	0.87	0.75	0.95	0.73	0.91	0.91	1.15
CRNN [17]	0.90	1.11	0.99	1.24	0.80	0.99	0.79	0.98	1.02	1.18
EEWNet [2]	0.90	1.07	1.01	1.18	1.02	1.18	0.81	0.98	0.99	1.15
CREIME [16]	**0.65**	0.82	0.74	0.91	0.71	0.89	0.78	0.95	0.97	1.14
MFTnet [18]	0.84	1.04	0.88	1.16	0.94	1.13	1.11	1.37	1.06	1.37
DNN regression [7]	1.43	3.54	1.71	3.01	1.59	3.41	1.56	3.21	2.01	4.35
MagNet [4]	0.77	0.95	0.83	1.01	0.88	1.07	0.96	1.15	0.99	1.15
DFTQuake [1]	**0.65**	**0.78**	**0.64**	**0.82**	0.72	0.88	0.71	0.89	0.70	0.87
MagPred [41]	0.88	0.98	0.87	0.95	0.92	1.03	0.94	1.05	0.92	1.08
Flowformer [25]	0.80	1.03	0.80	1.01	0.79	1.02	0.79	0.98	0.78	0.98
Reformer [24]	0.91	1.11	0.93	1.09	0.92	1.07	0.91	1.02	0.90	1.01
iTransformer [6]	1.23	1.56	1.19	1.46	1.10	1.32	1.08	1.21	1.09	1.12
Proposed method	0.69	0.86	0.67	0.84	**0.68**	**0.86**	**0.68**	**0.85**	**0.68**	**0.85**

DFTQuake [1] is almost 8.5 times higher than the proposed method as shown in Table 5. The second-best-performing model across five cases is the vision transformer model by Saad et al. (2024) [62]. However, the training time per epoch of the model by Saad et al. (2024) [62] is 23.47, 22.67, 23.56, 40.75, and 52.86 s, as shown in Table 5 due to the advanced network structure used by the model. Recently, deep random vector functional link (RVFL) neural networks have been widely applied to various tasks due to their simplicity and strong performance across different domains [63,64]. Therefore, in this study, we include the ensemble deep RVFL (edRVFL) [65] framework for comparison alongside ESN, as reported in Table 4.

The best time, shown by boldface, is obtained from the [7] method as shown in Table 5. The network by [7] utilizes a few layers of dense network only for prediction, which is also present as a DENN layer in the EarthESND model at the last layer. The reason for this increase in time is that the overhead of reservoir learning is present in the EarthESND network, which results in an increase in training time per epoch compared to the [7] method. The second-best time is obtained from the proposed model. It can be observed that for the [7] model, the training time per unit error is approximately 2.00, 2.49, 3.19, 2.61, and 2.64 s across the 2SE, 3SE, 4SE, 5SE, and 5SE datasets. In comparison, the EarthESND model achieves a much lower MAE of 0.78, 0.76, 0.77, 0.75, and 0.76

Table 5
The time (↓) in seconds taken per epoch for model training. The best timing is presented by a boldface.

Method	2SE	3SE	4SE	5SE	6SE
LSCross-ESN [35]	156.18	163.28	179.98	189.23	200.32
LS-ESN [34]	12.88	13.54	14.76	15.64	16.48
Inter-ESN [36]	40.13	49.94	55.90	50.68	65.15
ϕ-ESN [28]	5.46	7.53	7.89	8.43	10.64
Vanilla ESN [26]	11.23	16.45	15.28	18.78	21.84
Deep ESN [29]	22.41	25.45	29.64	31.16	34.87
Ensemble ESN [32]	20.61	26.21	32.75	36.86	33.12
edRVFl [65]	10.64	22.73	28.12	32.83	37.26
edESN [66]	24.32	29.56	32.64	36.32	39.16
SNRNN [9]	225.98	342.71	453.61	544.41	643.15
ViT magnitude [5]	23.47	22.67	23.56	40.75	52.86
CRNN [17]	24.79	30.86	25.90	39.07	45.23
EEWNet [2]	80.35	90.77	98.14	107.73	121.34
CREIME [16]	13.78	14.05	15.25	21.58	34.24
MFTnet [18]	25.49	21.39	21.01	42.79	45.98
DNN regression [7]	**4.74**	**5.92**	**5.34**	**5.16**	**6.86**
MagNet [4]	10.68	11.62	14.62	16.84	18.43
DFTQuake [1]	67.44	74.65	82.69	89.90	94.12
MagPred [41]	6.53	7.13	7.27	8.85	9.02
Flowformer [25]	31.09	30.42	32.77	32.48	33.34
Autoformer [22]	256.23	267.84	487.74	534.56	611.74
iTransformer [6]	25.98	29.76	30.43	33.88	34.87
Reformer [24]	32.41	36.88	42.65	46.89	50.21
Proposed method	7.80	6.71	10.55	9.32	10.99

M_{JMA} with comparable training times of 3.27, 6.72, 5.59, 8.19, and 7.04 s for respective datasets as shown in Table 5. This error demonstrates the ability of EarthESND to deliver significantly higher performance without a substantial increase in training time.

The higher errors in the other models are due to the removal of the epicenter distance limit in the current study, which introduces outliers that the previous methods are unable to handle effectively. *Supplementary Figure 2* shows a flowchart illustrating the differences between regression-based, deep learning (CNN [4,16], RNN [9,17], Transformer [5,18]), and reservoir computing network proposed in this work in EEWS. For transparency and reproducibility, a comprehensive summary of all model-specific hyperparameters is provided in the *Supplementary Table 11, 12, and 13*. *Supplementary Table 7* summarizes the model's single sample inference latency in milliseconds. We have added the number of learnable parameters for the state-of-the-art early magnitude prediction models in *Supplementary Table 8*, as learnable parameters directly contribute to the computational cost required during inference.

In addition to these models, we also evaluated the proposed method against widely used regression relationships in EEWS, discussed in detail in *Supplementary Section 2.7, Table 9*. The regression relations are designed considering the output in moment magnitude M_w scale as discussed in *Supplementary Section 2.7, Eq. (8)*. These regression-based approaches are compared with the EarthESND model as described by the mathematical formula in *Supplementary Table 9*. The cumulative absolute absement (CAA) parameter was used by Lin and Wu (2025) [3] and Wu et al. (2023) [67]. For the 6-second duration, the same equations used for the 5-second P-wave were adopted from Kumar et al. (2020) [68], Wu et al. (2023) [67], and Lin and Wu (2025) [3] due to the unavailability of specific 6-second P-wave equations. Similarly, for the 2SE dataset in Kumar et al. (2020) [68], the equations for the 3-second P-wave were used because no dedicated 2-second P-wave equations were available. Table 6 represents the quantitative analysis of the state-of-the-art regression relations for EEW magnitude prediction on the 2SE, 3SE, 4SE, 5SE, and 6SE datasets. It can be seen that the proposed method outperforms the other regression relations. The recent regression relation given by Lin and Wu (2025) [3] and Wu et al. (2023) [67] gives more error compared to the proposed method. This is because they are designed using different magnitude range of 4.5 to 7.5 M_w and 5.0 to 8.5 M_w as shown in *Supplementary Section 2.7, Table 9*.

4.3. Multi-station and inter-region study

2024 noto earthquake. The great Noto earthquake occurred on January 1, 2024, with a magnitude of 7.6 M_{JMA} in the Noto Peninsula. The area was severely affected, and the rescue operation was delayed [73]. The prior prediction of earthquake magnitude during an event will give lead time for rescuers to operate strategically. In this study, the approach was extended to multi-station magnitude prediction for the 2024 Noto earthquake, used as the test set with 232 records, as shown in Fig. 4. Fig. 4 represents the 2024 Noto event magnitude error with respect to epicenter distance using (a) 2, (b) 3, (c) 4, (d) 5, and (e) 6 s of P wave. Fig. 4 (f-j) represents the prediction error distribution with respect to epicenter distance as shown in the 2SE, 3SE, 4SE, 5SE, and 6SE datasets. Fig. 4 (k-o) S wave arrival time distribution with respect to the prediction error of 2SE, 3SE, 4SE, 5SE, and 6SE datasets. The quantitative analysis, based on the MAE and RMSE of the Noto event using several state-of-the-art methods, is shown in *Supplementary Section 2.8*.

Table 6
Comparison of state-of-the-art regression relations for early earthquake magnitude prediction of the test dataset. The boldface values represent the least error. The results are in M_w.

Parameter	Reference	Region	2SE		3SE		4SE		5SE		6SE	
			MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)	MAE(↓)	RMSE(↓)
τ_c	[69]	Japan	1.06	1.26	1.18	1.44	1.01	1.26	1.01	1.26	0.99	1.15
τ_p	[70]	USA	3.20	4.45	3.32	4.71	3.21	4.45	3.21	4.45	3.20	4.43
P_d	[70]	USA	3.76	3.89	3.46	3.60	3.76	3.89	3.76	3.89	3.78	3.89
τ_c	[71]	Taiwan	0.81	1.02	0.89	1.11	0.80	1.02	0.80	1.02	1.01	1.26
τ_c	[68]	India	4.87	5.12	5.34	5.60	4.87	5.12	4.74	5.11	4.87	5.12
P_d	[68]	India	0.91	1.14	0.81	1.01	0.91	1.14	0.90	1.12	0.91	1.14
τ_c	[72]	Japan, Taiwan, Italy	1.34	1.60	1.59	1.87	1.34	1.61	1.24	1.51	1.34	1.61
τ_c	[50]	Japan	1.43	1.69	1.71	1.99	1.43	1.69	1.23	1.48	1.18	1.32
CAA	[67]	Japan	4.41	4.52	4.63	4.77	4.85	4.91	4.99	5.07	4.56	4.98
P_d	[67]	Japan	1.10	1.36	1.15	1.42	1.10	1.36	1.10	1.36	0.99	1.21
CAA	[3]	Japan	4.51	5.69	4.54	5.32	4.51	5.68	4.21	5.67	4.12	5.23
P_d	[3]	Japan	3.21	4.65	3.93	4.57	3.43	4.21	3.54	4.12	3.12	3.87
Proposed method		Japan	**0.55**	**0.68**	**0.53**	**0.67**	**0.54**	**0.67**	**0.54**	**0.67**	**0.54**	**0.67**

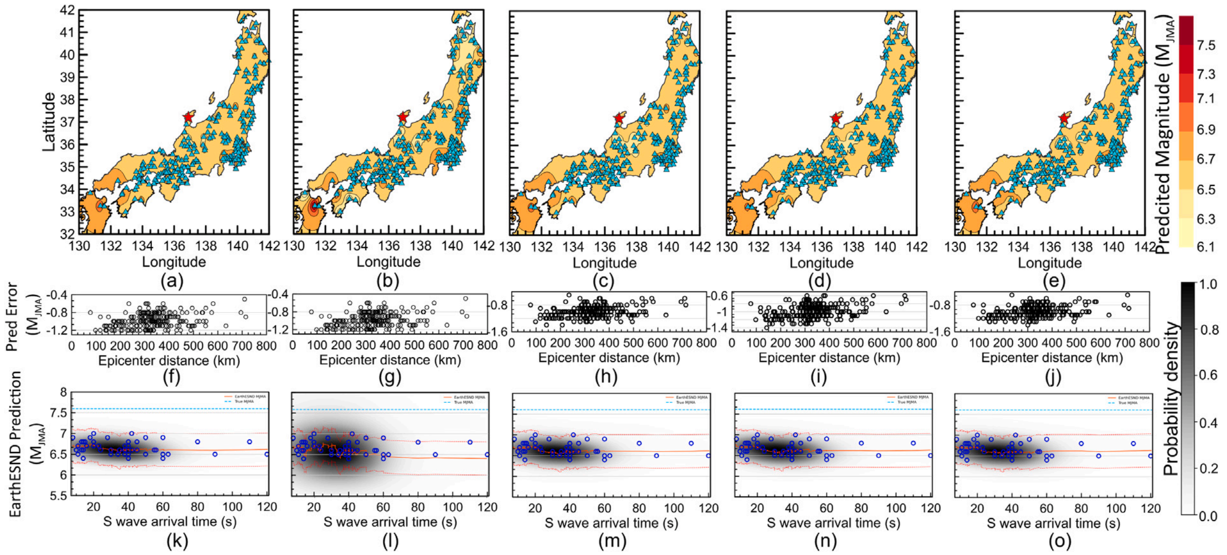

Fig. 4. The station-based contour maps of predicted magnitudes from the proposed model for the 7.6 M_{JMA} 2024 Noto event are shown for (a) 2 s, (b) 3 s, (c) 4 s, (d) 5 s, and (e) 6 s of P-wave data. Red star and cyan triangle denote the epicenter and station locations. The corresponding prediction errors (predicted - actual) with respect to epicentral distance are shown for (f) 2 s, (g) 3 s, (h) 4 s, (i) 5 s, and (j) 6 s of P-wave data. Similarly, the prediction errors (predicted - actual) with respect to time advantage before S-wave arrival are presented for (k) 2 s, (l) 3 s, (m) 4 s, (n) 5 s, and (o) 6 s of P-wave data. Plot (k–o) also includes the time magnitude density plot of EarthESND predictions for the 2024 Noto event. The red line shows the smoothed predicted magnitude, the red dashed line shows ±0.4 Mw uncertainty bounds, and the blue dashed line marks the true M_{JMA} magnitude. 232 records of 2024 Noto events are selected for this study.

Inter region study: India. Cross-region testing was conducted on the Indian dataset to assess the model's performance on inter-regional data. The MAE of 1.00, 0.97, 1.21, 1.24, and 1.11 M_w have been obtained from the 2, 3, 4, 5, and 6 s of P wave of the Indian dataset. The RMSE values of 1.13, 1.13, 1.32, 1.35, and 1.27 M_w were obtained from the 2, 3, 4, 5, and 6 s of the P wave in the Indian dataset, which comprised 157 records. Fig. 5 (a-e) represents the actual and predicted magnitude of the five datasets. The Fig. 5 (f-j) represents the error distribution with respect to epicenter distance (km) for five datasets. The high prediction error in the Indian region results from training the model in a region of Japan with different geography and testing it in another area.

5. Conclusion

The proposed work develops a framework that utilizes ESN to predict the onsite EEW magnitude. The proposed framework is scalable due to its significantly lower training time compared to advance deep learning models, and multiscale as it captures dynamics across multiple temporal scales within the ESN reservoir. The ablation study and parameter analysis comprehensively validate the performance of the EarthESND model compared to its variants. The proposed method shows better performance with respect to recent deep learning networks that additionally require longer training time, as shown in Table 4. Moreover, It can

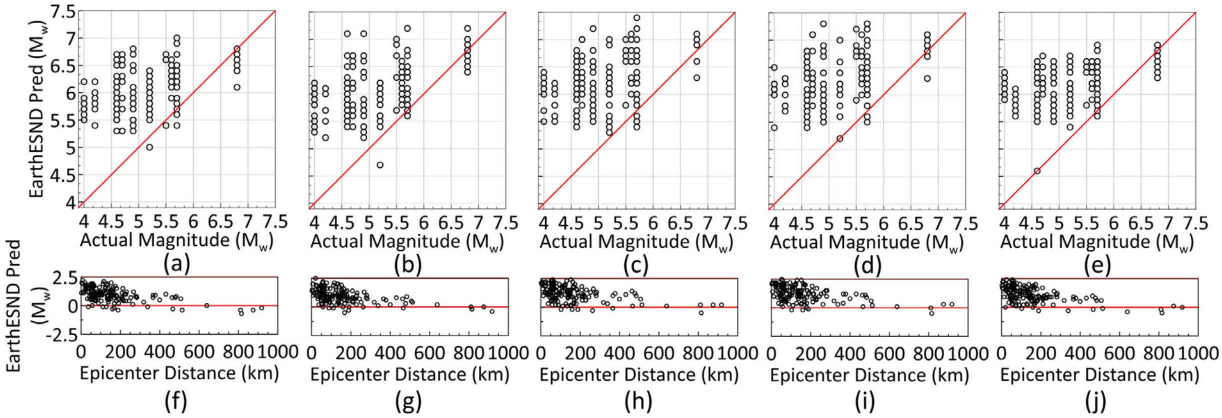

Fig. 5. The station-based contour maps of predicted magnitudes from the proposed model for the India earthquake are shown for (a) 2 s, (b) 3 s, (c) 4 s, (d) 5 s, and (e) 6 s of P-wave data. The corresponding prediction errors (predicted - actual) with respect to epicentral distance (km) are shown for (f) 2 s, (g) 3 s, (h) 4 s, (i) 5 s, and (j) 6 s of P-wave data. The dataset consists of 157 records of 12 earthquakes.

be observed from Table 6 that the proposed EarthESND network outperforms the recently and widely used regression relations for earthquake prediction. The research findings of this study establish an advanced EEW framework that relies on high-end computational systems for magnitude prediction. The geoscience and computer science communities can apply the proposed model to identify and monitor severely impacted regions during an earthquake, thereby helping reduce potential damage and losses at low computational cost. Future work can evaluate the model's robustness under low-SNR conditions to assess stability in noisy environments and can further explore the EEW magnitude detection using broadband sensors. Another future direction is the development of a framework for identifying noise in earthquake signals, which may also support more reliable P-wave onset detection.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compeleceng.2026.111161.

Data and Code Availability

The source codes are available for download at the link: https://github.com/anushka-joshi/EarthESND-Model. Event datasets were curated from Japan's Kyoshin Net (K-NET) mained by National Research Institute for Earth Science and Disaster Resilience (NIED) [59] and India's PESMOS that is a project run by department of earthquake engineering, Indian Institute of Technology Roorkee, Roorkee (https://pesmos.org). We thank the data providers for public access.

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